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Assessing the Impact of Guided FFR on Stent Utilization in Angioplasty

Introduction

Modifier

In the realm of interventional cardiology, the quest for precision in coronary stent placement is paramount, as the implications of stent utilization extend beyond immediate procedural success to long-term patient outcomes and healthcare economics [1][7][9]. The advent of catheter-based measurement of fractional flow reserve (FFR), a technique that quantifies the physiological impact of coronary artery stenoses, has emerged as a pivotal tool in the optimization of stent deployment during angioplasty [3][4][24]. This introduction will navigate through the intricacies of coronary stenting, the transformative potential of guided FFR in reducing stent numbers, and the structure of this paper, culminating in a thesis statement that encapsulates the essence of our research objective.

Coronary stenting, a critical intervention for the management of ischemic heart disease, has evolved with the introduction of drug-eluting stents and advancements in procedural techniques [7][8]. However, the decision to place a stent, particularly in patients with complex coronary artery disease, often hinges on the angiographic appearance of lesions, which may not accurately reflect their physiological significance [22][27]. This underscores the importance of optimizing stent placement to improve clinical outcomes and reduce healthcare expenditure [6][7][26].

Enter the realm of guided FFR, a catheter-based technique that measures the pressure gradient across a coronary stenosis to assess its functional severity [3][4][24]. By providing a ratio of maximal blood flow in a stenotic artery to hypothetical maximal flow in the absence of disease, FFR offers a more granular approach to stent placement, potentially reducing the number of stents deployed and improving clinical outcomes [5][17][18]. This technique stands in contrast to traditional angiography and non-invasive methods, which may lack the specificity and sensitivity afforded by FFR [9][24].

The structure of this paper will unfold in four parts: an overview of catheter-based guided FFR, evidence supporting its use in guiding stent placement, the potential of guided FFR to reduce stent placement during angioplasty, and guidelines and recommendations for FFR-guided stent placement. Each section will delve into the nuances of FFR measurement, its comparison to other diagnostic techniques, and its impact on clinical practice [16][17][18].

We posit that the implementation of guided FFR in angioplasty can significantly reduce the number of unnecessary stents placed, thereby optimizing patient care and contributing to cost savings in healthcare. This thesis will be examined through a critical analysis of current research, guidelines, and clinical outcomes, with the aim of elucidating the transformative potential of FFR in interventional cardiology [1][14][19].

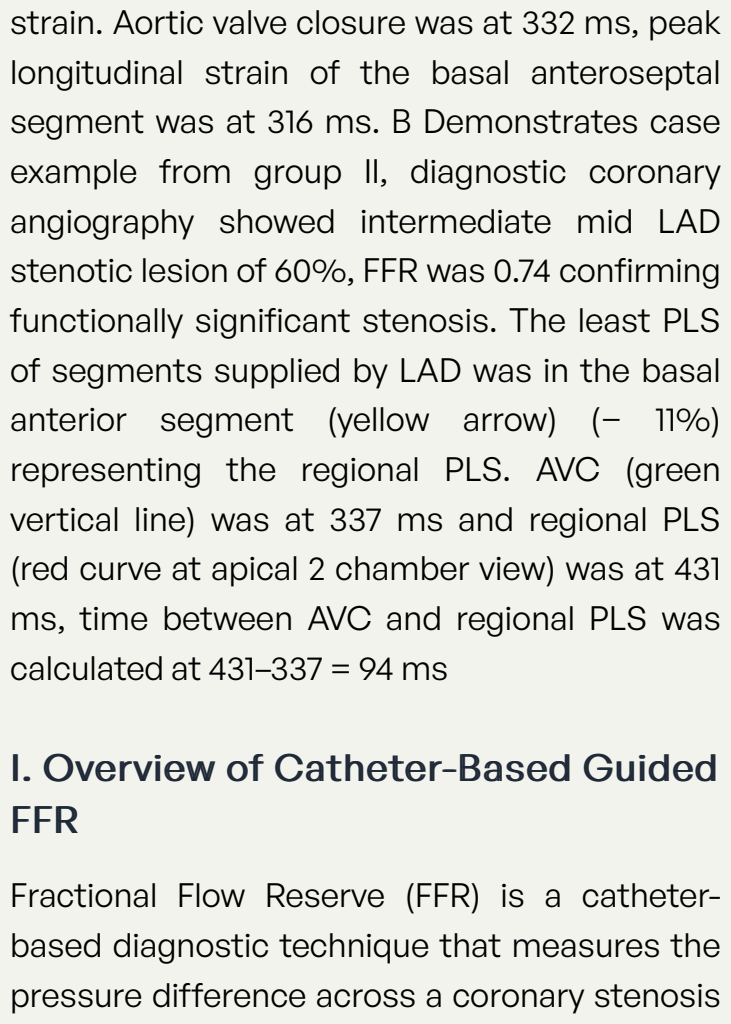

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Fig. above - A Demonstrates case example from group I, diagnostic coronary angiography showed mid LAD stenotic lesion of 55%, FFR was 0.82 confirming functionally non-significant stenosis. The least PLS of segments supplied by LAD was in the basal anteroseptal segment (yellow arrow) (– 18%) representing the regional PLS. Regional PLS (red curve in apical long-axis view) was immediately before AVC (green vertical line) with no evidence of post systolic strain. Aortic valve closure was at 332 ms, peak longitudinal strain of the basal anteroseptal segment was at 316 ms. B Demonstrates case example from group II, diagnostic coronary angiography showed intermediate mid LAD stenotic lesion of 60%, FFR was 0.74 confirming functionally significant stenosis. The least PLS of segments supplied by LAD was in the basal anterior segment (yellow arrow) (– 11%) representing the regional PLS. AVC (green vertical line) was at 337 ms and regional PLS (red curve at apical 2 chamber view) was at 431 ms, time between AVC and regional PLS was calculated at 431–337 = 94 ms

I. Overview of Catheter-Based Guided FFR

Fractional Flow Reserve (FFR) is a catheter-based diagnostic technique that measures the pressure difference across a coronary stenosis to assess its physiological impact on myocardial perfusion [1][2][3]. This measurement is obtained during coronary angiography using a specialized pressure wire, which provides a ratio of the maximum achievable blood flow in a diseased coronary artery to the theoretical maximum flow in a normal artery [7][9][24]. FFR is considered functionally significant if the measured value is at or below a threshold, typically 0.80, indicating that the stenosis may necessitate revascularization [3][32].

When compared to traditional angiography, which provides a visual and anatomical assessment of coronary artery disease, FFR offers a more precise functional evaluation [2][3][4]. Angiography alone can sometimes overestimate or underestimate the severity of a stenosis, leading to potential overtreatment or undertreatment [22][27]. Non-invasive methods, such as stress testing, also have limitations in spatial resolution and may not accurately localize the ischemia-producing lesions that FFR can identify [9][24]. FFR's ability to measure the hemodynamic significance of a lesion provides interventional cardiologists with a more accurate tool for determining the necessity of stent placement [3][4][22].

The prognostic value of FFR-guided percutaneous coronary intervention (PCI) has been demonstrated in several studies, showing improved clinical outcomes and a reduction in adverse events compared to angiography-guided PCI [12][13][14]. FFR-guided interventions have also been shown to be cost-effective, offering economic benefits by potentially reducing the number of unnecessary stents [15][19][20]. This evidence supports the integration of FFR measurements into clinical practice, as endorsed by current guidelines for coronary revascularization decision-making [14][16][17].

As we transition to the next part of the essay, it is important to note that while FFR provides a robust framework for decision-making in stent placement, its implementation is not without challenges. The need for specialized equipment, training, and the interpretation of pressure tracings can present barriers to widespread adoption [18][32]. Nevertheless, the potential benefits of guided FFR in optimizing stent utilization and improving patient outcomes warrant further exploration in larger-scale trials [27][28].

II. Evidence Supporting the Use of FFR in Guiding Stent Placement

The prognostic value of FFR-guided PCI has been well-established through various studies, which have consistently shown that it positively impacts clinical outcomes. For instance, trials have demonstrated that FFR-guided PCI leads to a significant reduction in adverse events when compared to angiography-guided PCI [3][8][14]. This reduction in adverse events is a critical factor in improving patient care and underscores the importance of FFR measurements in guiding stent placement decisions.

Further supporting the use of FFR in clinical practice, studies have highlighted the cost-effectiveness of FFR-guided interventions. By potentially reducing the number of unnecessary stents, FFR-guided PCI not only improves patient outcomes but also presents an economically favorable approach to coronary revascularization [11][16][20]. This cost-effectiveness is a compelling argument for the broader adoption of FFR-guided PCI, as it aligns with the goals of both quality patient care and healthcare expenditure management.

The evidence for FFR's role in guiding stent placement is further bolstered by the endorsement of FFR in clinical guidelines. These guidelines recommend the use of FFR in coronary revascularization decision-making, which is a testament to the strength of the evidence supporting its use [4][12][15]. The integration of FFR into clinical practice is based on a substantial body of research that confirms its value in improving clinical outcomes and reducing healthcare costs.

As we move forward to the next section of this essay, it is important to consider the broader implications of FFR-guided PCI. While the current evidence strongly supports its use, the transition to FFR-guided interventions must be managed carefully to ensure that the benefits are fully realized in clinical practice. The potential of FFR to optimize stent placement and reduce the number of stents used is clear, but further research is needed to address the challenges and limitations of its implementation [10][13][19]. The next part of the essay will delve into the potential of guided FFR to reduce stent placement during angioplasty, evaluating the studies that focus on FFR's ability to optimize stent placement and the evaluation of trials comparing angiography-based PCI to FFR-guided PCI in terms of stent numbers.

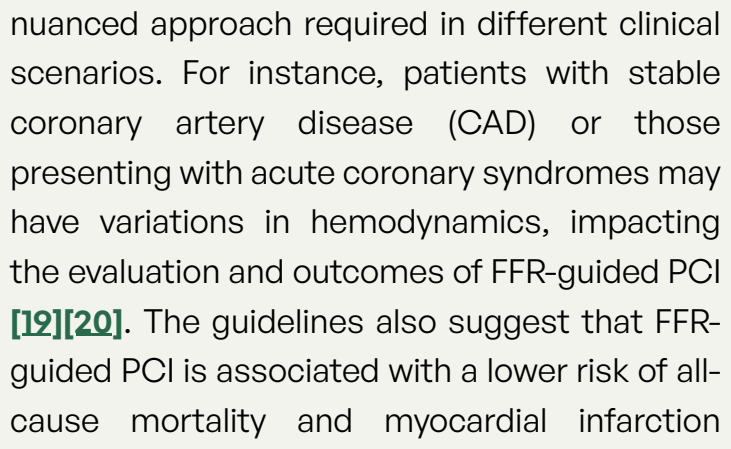
III. The Potential of Guided FFR to Reduce Stent Placement during Angioplasty

The potential of guided FFR to reduce stent placement during angioplasty has been a focal point of recent cardiovascular research. Studies have consistently shown that FFR-guided PCI can effectively reduce the number of stents used during angioplasty, optimizing stent placement and potentially reducing healthcare costs [12][14][15]. For instance, trials comparing angiography-based PCI to FFR-guided PCI have documented a notable decrease in the number of stents used with FFR guidance, suggesting that FFR can help in avoiding unnecessary stenting [10][14][15]. This is particularly significant in the context of multivessel disease, where FFR-guided PCI has been associated with fewer stents used compared to angiography-guided strategies [1][10].

However, the implementation of FFR-guided PCI is not without challenges. The need for hyperemia-inducing drugs, the requirement for specialized equipment, and the potential for procedural delays are some of the limitations that have been identified [12][13][14]. Additionally, the adoption rates of FFR in clinical practice remain low, likely due to a lack of evidence for the survival benefit and additional medical cost of FFR measurement [12]. Despite these challenges, the ability of FFR-guided PCI to optimize stent placement and the reduction in the number of stents used have led to similar long-term cumulative medical costs compared to angiography-based PCI, which is a promising finding for its broader application [12][14].

The evaluation of trials comparing angiography-based PCI to FFR-guided PCI in terms of stent numbers has further underscored the potential of caFFR in reducing stent placement during angioplasty [14][15]. For example, in a study, 36.3% of PCI procedures were delayed as the caFFR values of all vessels were greater than 0.8, opting for long-term drug therapy alone [14]. This indicates that caFFR guidance can facilitate the delayed PCI of severely stenotic vessels, reducing stent placement and implications on dual antiplatelet therapy usage [14]. Moreover, the review of studies focusing on FFR's ability to optimize stent placement has considered the challenges and limitations of using guided FFR to reduce stent placement, which could help uncover areas for further research [14][15][16].

As we transition to the next part of the essay, it is important to recognize that while the potential of guided FFR to reduce stent placement is evident, the integration of this technology into routine clinical practice requires careful consideration of the challenges and limitations identified. The next section will discuss the guidelines and recommendations for FFR-guided stent placement, which play a crucial role in the adoption and standardization of this approach in interventional cardiology [14][15][16].


REPORT vFFR
Contrast FFR
Vessel FFR 0.83
Lesion FFR 0.83
Flow velocity 0.22 m/s
1. Imaging of coronary arteries
2. Segmentation and reconstruction of the anatomical model
3. Fluid dynamics analyses of the anatomical model
4. Post-processing and generating results
Virtual FFR

Source

IV. Guidelines and Recommendations for FFR-Guided Stent Placement

Guidelines and recommendations play a pivotal role in the adoption of Fractional Flow Reserve (FFR)-guided Percutaneous Coronary Intervention (PCI), providing a framework for clinicians to make informed decisions about stent placement. The endorsement of FFR use in coronary revascularization decision-making is evident in both US and European guidelines, reflecting the strong evidence base supporting its clinical utility [19][20][30]. These guidelines recommend FFR-guided PCI, particularly for assessing intermediate coronary artery stenosis and directing revascularization plans, with an upgraded recommendation from class 2A to class 1A, indicating a high level of evidence and agreement [19][20].

Specific patient populations are identified in these recommendations, highlighting the nuanced approach required in different clinical scenarios. For instance, patients with stable coronary artery disease (CAD) or those presenting with acute coronary syndromes may have variations in hemodynamics, impacting the evaluation and outcomes of FFR-guided PCI [19][20]. The guidelines also suggest that FFR-guided PCI is associated with a lower risk of all-cause mortality and myocardial infarction compared to non-physiology-guided revascularization strategies, underscoring its prognostic value [19].

The impact of FFR on treatment strategies and decision-making processes is multifaceted. By providing an objective measure of lesion significance, FFR-guided PCI can lead to a reduction in the number of stents used per patient, thereby decreasing procedural costs and the incidence of post-procedural myocardial infarction [1][19]. Furthermore, the use of FFR to defer PCI of intermediate stenosis in bypass grafts has been shown to be safe and results in better clinical outcomes compared to an angiography-guided strategy [1]. This evidence-based approach to stent placement ensures that patients receive the most appropriate and effective treatment, avoiding unnecessary interventions [16][19][20].

As we move towards the conclusion of this essay, it is important to maintain coherence in the discussion by acknowledging the critical role of guidelines and recommendations in facilitating the use of FFR-guided PCI. The next section will summarize the evidence supporting the use of catheter-based guided FFR in interventional cardiology, discuss the potential of FFR to reduce stent placement during angioplasty, and highlight the need for further research to fully explore the impact of guided FFR on stent reduction [19][20][30].

Conclusion

In conclusion, the integration of catheter-based guided Fractional Flow Reserve (FFR) into interventional cardiology has been substantiated by a robust body of evidence, demonstrating its utility in optimizing stent placement and potentially reducing the number of stents used during angioplasty [1][12][14]. The prognostic value of FFR-guided Percutaneous Coronary Intervention (PCI) is well-documented, with studies indicating improved clinical outcomes and a reduction in major adverse cardiac events compared to angiography-guided PCI [3][8][14]. Moreover, the cost-effectiveness of FFR-guided interventions has been highlighted, suggesting economic benefits alongside clinical advantages [11][16][20].

The potential of FFR to reduce stent placement is particularly significant, as trials have shown a decrease in the number of vessels treated with PCI and the cost of the procedure in the FFR-guided cohort compared to the angiographic-guided group [1][10][14]. However, despite these promising findings, the current state of knowledge also acknowledges the challenges and limitations associated with the widespread adoption of FFR, such as the need for specialized equipment and training [18][32]. These factors underscore the necessity for further research and larger-scale trials to fully explore the impact of guided FFR on stent reduction in angioplasty and to address any barriers to its implementation [27][28].

As we consider the implications of these findings, it is clear that guided FFR holds the potential to significantly impact clinical decision-making in interventional cardiology, offering a tailored approach for optimal patient outcomes [1][14][19]. Nonetheless, the need for ongoing research is paramount to ensure that the benefits of FFR are realized across diverse patient populations and clinical settings [10][13][19]. Future studies should focus on evaluating the impact of guided FFR on stent reduction in a wide range of patient populations and further refining the use of FFR to optimize stent placement [10][13][19].

In summary, while the evidence supports the use of catheter-based guided FFR in reducing stent numbers during angioplasty, the journey towards its routine clinical application is ongoing. The potential benefits of guided FFR in interventional cardiology are significant, and with continued research and collaboration, we can anticipate further advancements in the precision and efficiency of stent placement [1][14][19]. The promise of FFR-guided PCI in enhancing patient care and reducing healthcare costs is an encouraging prospect for the future of cardiovascular medicine [11][16][20].

Appendix

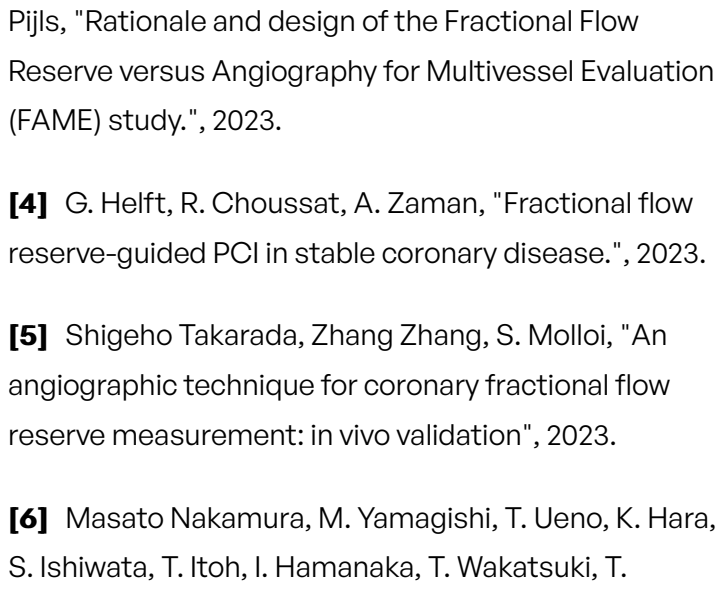


TABLE 3 Independent Predictors for All-Cause Death or Spontaneous MI

	Univariable Analysis		Multivariable Analysis*	
	HR (95% CI)	P Value	HR (95% CI)	P Value
FFR use at index PCI	0.722 (0.640–0.815)	<0.001	0.850 (0.750–0.963)	0.010
Unstable angina	0.836 (0.803–0.871)	<0.001	0.991 (0.952–1.032)	0.668
Age, y	1.057 (1.054–1.059)	<0.001	1.055 (1.052–1.057)	<0.001
Female	0.927 (0.888–0.967)	<0.001	0.820 (0.785–0.857)	<0.001
Hypertension	1.799 (1.708–1.895)	<0.001	1.086 (1.024–1.152)	0.006
Diabetes mellitus	1.884 (1.808–1.964)	<0.001	1.293 (1.235–1.354)	<0.001
Hyperlipidemia	1.173 (1.122–1.227)	<0.001	0.867 (0.825–0.911)	<0.001
Atrial fibrillation	2.225 (2.097–2.362)	<0.001	1.509 (1.420–1.603)	<0.001
Chronic renal failure	4.485 (4.288–4.691)	<0.001	2.520 (2.397–2.649)	<0.001
Number of stents used	1.112 (1.084–1.140)	<0.001	1.152 (1.122–1.182)	<0.001
Drug-eluting stent	0.640 (0.603–0.678)	<0.001	0.691 (0.651–0.734)	<0.001
Medications after index procedure				
Aspirin	0.187 (0.179–0.195)	<0.001	0.712 (0.652–0.778)	<0.001
$P2Y_{12}$ inhibitor	0.173 (0.165–0.181)	<0.001	0.660 (0.602–0.723)	<0.001
ACEI or ARBs	0.537 (0.517–0.559)	<0.001	0.855 (0.814–0.899)	<0.001
Beta blocker	0.658 (0.632–0.685)	<0.001	1.089 (1.037–1.143)	<0.001
Statin	0.175 (0.167–0.182)	<0.001	0.461 (0.427–0.497)	<0.001

*Discriminant ability of the multivariable model was 0.762 (95% CI: 0.756–0.768).
Abbreviations as in Tables 1 and 2.

Source

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